

Hierarchical Bayesian Decomposition of Football Player Performance in La Liga

A Factor-Structured Additive Model with Low-Rank Player Effects and Diagonal Season Effects

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Table of contents

1	Introduction	3
1.1	Football Analytics and the Measurement of Player Quality	3
1.2	Objectives	3
1.3	Contribution of This Work	3
2	Data	4
2.1	Source	4
2.2	Longitudinal Aggregation	4
2.3	Eligibility Filter	4
2.4	Final Dataset Dimensions	4
2.5	Feature Engineering	4
2.6	Feature Scaling	5
3	Mathematical Model Specification	6
3.1	Notation	6
3.2	Additive Decomposition	6
3.3	Model A: Diagonal Baseline (Stages 03, 05)	6
3.4	Model B: Full Low-Rank Recovery (Stage 06) — Failed	7
3.5	Model D: Low-Rank Σ_a + Diagonal Σ_b (Stages 09, 10)	7
3.5.1	Factor Model Interpretation	7
3.5.2	Variance Decomposition	8
4	Identifiability and Priors	9
4.1	The Identifiability Problem in Factor Models	9
4.2	Anchor Constraints (Pure-Variable Restriction)	9
4.3	Non-Centred Parameterisation	9
4.4	Prior Distributions	10
5	Bayesian Inference with NUTS	11
5.1	Hamiltonian Monte Carlo	11
5.2	The No-U-Turn Sampler	11

5.3	Convergence Diagnostics	11
6	Results: Variance Decomposition	13
6.1	Main Result — Diagonal Model (Stage 03)	13
6.2	Interpretation by Feature Category	14
7	Simulation Study	16
7.1	Purpose	16
7.2	Data-Generating Process	16
7.3	Stage 05: Diagonal Model Robustness	16
7.4	Stage 06: Full Low-Rank Recovery — Negative Result	17
8	New Model: Low-Rank Σ_a + Diagonal Σ_b	18
8.1	Motivation	18
8.2	Model Specification	18
8.3	Posterior Health — Simulated Data Fit	18
8.4	Recovery of Λ_a — Simulated Data	18
8.5	Loading Table — Real Data Fit	19
9	Loading Interpretation and Football Analysis	20
9.1	Reading Factor Loadings	20
9.2	Group-Level Analysis	20
9.3	Player Archetypes	20
9.4	Comparison of Loading Patterns	20
10	Discussion	21
10.1	What Worked	21
10.2	What Failed and Why	21
10.3	Season Effect Interpretation	21
11	Future Improvements	22
11.1	1. K-Fold Cross-Validation	22
11.2	2. More Seasons / Multiple Leagues	22
11.3	3. Position-Specific Models	22
11.4	4. Spike-and-Slab Priors on Loadings	22
11.5	5. Dynamic Season Effects	22
11.6	6. Player Trajectory Model	22
12	Appendix A: Mathematical Derivations	23
12.1	A.1 Marginal Likelihood of the Factor Model	23
12.2	A.2 Non-Centred Parameterisation Jacobian	23
12.3	A.3 Column-Centering and Identifiability of μ	23
12.4	A.4 Woodbury Identity and Fast Likelihood Computation	24
13	Appendix B: Computational Details	25
13.1	B.1 CmdStan Configuration	25
13.2	B.2 Memory Management	25
13.3	B.3 Stan Model Files	25
13.4	B.4 Reproducibility	25

1 Introduction

1.1 Football Analytics and the Measurement of Player Quality

Modern professional football generates an enormous volume of event-level data. Tracking every pass, shot, dribble, recovery, and tackle provides a rich substrate for quantitative analysis of player performance. Yet raw event counts are confounded by many factors: a player accumulates more recoveries if they play a pressing style; a forward records more shots if their team dominates possession; success rates are distorted by small sample sizes.

A principled statistical approach must separate *stable player attributes* from *contextual variation* (season, team style, league difficulty) and *random noise*. This document describes a hierarchical Bayesian additive model that does exactly this: it decomposes each performance feature into a player effect, a season effect, and residual noise — and, in its richest form, structures the player effect as a low-rank factor model to discover latent player archetypes.

1.2 Objectives

1. Quantify how much of the variation in 53 performance features is attributable to stable player quality vs. seasonal context vs. noise.
2. Model the **covariance structure** of player effects using a factor model $\Sigma_a = \Lambda_a \Lambda_a^\top + \Psi_a$, discovering axes of correlated player style.
3. Validate the model on simulated data with known ground truth.
4. Interpret the discovered factors in football terms.

1.3 Contribution of This Work

This thesis makes the following methodological contributions:

- A **fully Bayesian** treatment of the variance decomposition problem using No-U-Turn Sampler (NUTS) Hamiltonian Monte Carlo via Stan.
- A **non-centred parameterisation** that avoids funnel pathologies in the posterior geometry.
- An **anchor-constrained factor model** for the player covariance that resolves sign and permutation indeterminacy.
- A **simulation study** that validates recovery of the factor structure under realistic data dimensions.

2 Data

2.1 Source

The raw data are player-match aggregates from the **Wyscout** football event database, covering La Liga (Spanish top division). Each row in the raw file records a player’s performance statistics in a single match.

2.2 Longitudinal Aggregation

Match-level rows are aggregated to **player-season** observations. For each player i and season j , all matches in which the player participated with at least a minimum number of minutes are summed. Per-90 rates are computed as:

$$\text{per90}_{ij,k} = \frac{\text{count}_{ij,k}}{\text{minutes}_{ij}} \times 90$$

Success rates (e.g. pass accuracy) are computed as:

$$\text{rate}_{ij,k} = \frac{\text{successes}_{ij,k}}{\text{attempts}_{ij,k}}$$

Goalkeeper rows are excluded throughout.

2.3 Eligibility Filter

Only player-season rows with **at least 450 minutes** played are retained. This ensures that per-90 rates are based on sufficient playing time and that extreme rate estimates (1 successful tackle from 1 attempt) are not included.

2.4 Final Dataset Dimensions

Table 1: Dataset dimensions after filtering

item	value
N_player_seasons	4944
P_variables	53
N_players	1650
N_seasons	12
one_season_players	605
repeated_players	1045
players_with_gaps	346
feature_matrix_rank	53
feature_matrix_max_abs_correlation	1

2.5 Feature Engineering

53 features are selected and retained after auditing for:

- Zero or near-zero variance (excluded)
- Missing values: features with $> 20\%$ missing are excluded; remaining missing values are filled by per-feature median imputation.

The 53 features are grouped into eight football-meaningful categories:

Group	Example features
Attacking output	goals, shots, xG, touch in box, assists
Shooting rates	shots on target rate, goal conversion rate
Passing volume	passes, key passes, progressive passes, crosses
Passing quality	pass success rate, progressive pass success rate
Progressive actions	dribbles, progressive runs, accelerations
Defending	recoveries, interceptions, clearances, losses
Dueling	defensive/offensive/aerial/pressing duels, success rates
Discipline	fouls, fouls suffered, yellow cards

2.6 Feature Scaling

All 53 features are **standardised** (Z-scored) to zero mean and unit variance across all player-season rows:

$$\tilde{y}_{nk} = \frac{y_{nk} - \bar{y}_k}{s_k}$$

where $\bar{y}_k$ and s_k are the empirical mean and standard deviation of feature k computed over all N rows. Scaling ensures that all features contribute equally to the likelihood regardless of their original unit, and that prior scales are comparable across features.

3 Mathematical Model Specification

3.1 Notation

Symbol	Meaning
N	Number of player-season rows
P	Number of scaled features ($P = 53$)
I	Number of unique players ($I = 1,650$)
S	Number of seasons ($S = 12$)
Q_a	Player factor rank ($Q_a = 2$)
$i[n]$	Player index for row n
$j[n]$	Season index for row n

3.2 Additive Decomposition

All models share the same **mean structure**:

$$\mathbf{y}_n = \boldsymbol{\mu} + \mathbf{a}_{i[n]} + \mathbf{b}_{j[n]} + \boldsymbol{\varepsilon}_n, \quad n = 1, \dots, N$$

where:

- $\mathbf{y}_n \in \mathbb{R}^P$ is the vector of P scaled performance features for row n
- $\boldsymbol{\mu} \in \mathbb{R}^P$ is the overall population mean vector
- $\mathbf{a}_i \in \mathbb{R}^P$ is the **player effect** for player i , capturing stable attributes
- $\mathbf{b}_j \in \mathbb{R}^P$ is the **season effect** for season j , capturing temporal drift
- $\boldsymbol{\varepsilon}_n \in \mathbb{R}^P$ is the **residual** for row n , capturing row-level noise

The four models in this project differ only in the **covariance structure** imposed on $\mathbf{a}_i$, $\mathbf{b}_j$, and $\boldsymbol{\varepsilon}_n$.

3.3 Model A: Diagonal Baseline (Stages 03, 05)

The simplest specification assumes independent (diagonal) covariance for all components:

$$\begin{aligned} \mathbf{a}_i &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_a), & \boldsymbol{\Sigma}_a &= \text{diag}(\sigma_{a,1}^2, \dots, \sigma_{a,P}^2) \\ \mathbf{b}_j &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_b), & \boldsymbol{\Sigma}_b &= \text{diag}(\sigma_{b,1}^2, \dots, \sigma_{b,P}^2) \\ \boldsymbol{\varepsilon}_n &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon), & \boldsymbol{\Sigma}_\varepsilon &= \text{diag}(\sigma_{\varepsilon,1}^2, \dots, \sigma_{\varepsilon,P}^2) \end{aligned}$$

This model has $3P = 159$ variance parameters plus $P = 53$ mean parameters. Each feature is modelled **independently**: the fact that a player who scores many goals also takes many shots is not captured by the player covariance — it emerges only from the marginal distributions.

The diagonal assumption is a deliberate simplification that makes the model identifiable, computationally tractable, and interpretable via variance decomposition ratios.

3.4 Model B: Full Low-Rank Recovery (Stage 06) — Failed

A richer model attempts to capture cross-feature correlations in **both** player and season effects using factor (low-rank) covariances:

$$\begin{aligned}\Sigma_a &= \Lambda_a \Lambda_a^\top + \Psi_a, & \Lambda_a &\in \mathbb{R}^{P \times Q_a}, & \Psi_a &= \text{diag}(\psi_{a,1}^2, \dots, \psi_{a,P}^2) \\ \Sigma_b &= \Lambda_b \Lambda_b^\top + \Psi_b, & \Lambda_b &\in \mathbb{R}^{P \times Q_b}\end{aligned}$$

This is the standard **factor analytic** (or confirmatory factor analysis) covariance. The diagonal Ψ_a is called the **uniqueness** matrix; each element $\psi_{a,p}^2$ captures the variance of feature p not explained by the shared factors.

Why this model failed: With only $S = 12$ seasons, the season factor loadings Λ_b are severely under-identified. The sampler found many near-equivalent solutions and chains did not mix (maximum $\hat{R} = 1.739$, minimum bulk ESS = 6). The model is statistically correct but not estimable with this data.

3.5 Model D: Low-Rank Σ_a + Diagonal Σ_b (Stages 09, 10)

The new model, which is the central contribution of this thesis, makes a principled asymmetric choice:

$$\begin{aligned}\Sigma_a &= \Lambda_a \Lambda_a^\top + \Psi_a \\ \Sigma_b &= \text{diag}(\sigma_{b,1}^2, \dots, \sigma_{b,P}^2)\end{aligned}$$

Why this asymmetry makes sense:

- With $I = 1,650$ players, the player factor model is well-supported by the data. We have enough observations per factor to estimate $P \times Q_a = 106$ loading parameters reliably.
- With $S = 12$ seasons, we cannot reliably estimate a $P \times Q_b$ season loading matrix. Keeping Σ_b diagonal avoids the identifiability problems of Model B while still capturing feature-specific seasonal drift.

3.5.1 Factor Model Interpretation

The factor model for Σ_a can be written in terms of latent variables:

$$\mathbf{a}_i = \Lambda_a \boldsymbol{\eta}_i + \Psi_a^{1/2} \boldsymbol{\xi}_i$$

where:

- $\boldsymbol{\eta}_i \in \mathbb{R}^{Q_a}$ is the vector of **latent factor scores** for player i , with $\boldsymbol{\eta}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{Q_a})$
- $\boldsymbol{\xi}_i \in \mathbb{R}^P$ is the **uniqueness residual**, with $\boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$
- $\Lambda_a \in \mathbb{R}^{P \times Q_a}$ is the **loading matrix**: $\Lambda_{a,pq}$ measures how strongly feature p loads onto factor q

The sign of $\Lambda_{a,pq}$ indicates direction: a positive loading means that a player with a high factor score $\eta_{i,q}$ tends to have an above-average value on feature p , and vice versa.

The implied covariance is:

$$\text{Cov}(\mathbf{a}_i) = \mathbf{\Lambda}_a \mathbf{\Lambda}_a^\top + \mathbf{\Psi}_a = \mathbf{\Sigma}_a$$

3.5.2 Variance Decomposition

For each feature p , the total variance decomposes as:

$$\text{Var}(y_{np}) = \underbrace{\Sigma_{a,pp}}_{\text{player}} + \underbrace{\sigma_{b,p}^2}_{\text{season}} + \underbrace{\sigma_{\varepsilon,p}^2}_{\text{residual}}$$

with proportions:

$$\pi_{a,p} = \frac{\Sigma_{a,pp}}{\Sigma_{a,pp} + \sigma_{b,p}^2 + \sigma_{\varepsilon,p}^2}, \quad \pi_{b,p} = \frac{\sigma_{b,p}^2}{.}, \quad \pi_{\varepsilon,p} = \frac{\sigma_{\varepsilon,p}^2}{.}$$

where $\Sigma_{a,pp} = \sum_{q=1}^{Q_a} \Lambda_{a,pq}^2 + \psi_{a,p}^2$.

4 Identifiability and Priors

4.1 The Identifiability Problem in Factor Models

Factor models suffer from two fundamental indeterminacies that must be resolved for Bayesian inference to be well-posed:

- 1. Sign indeterminacy:** If $(\Lambda_a, \boldsymbol{\eta}_i)$ is a solution, so is $(-\Lambda_a, -\boldsymbol{\eta}_i)$. Multiplying both by -1 leaves the model likelihood unchanged. This creates a bimodal posterior.
- 2. Permutation indeterminacy:** Any permutation of the factors $(\Lambda_a \mathbf{P}, \mathbf{P}^\top \boldsymbol{\eta}_i)$ for a permutation matrix $\mathbf{P}$ is an equally valid solution. With $Q_a = 2$ factors, this gives $2! = 2$ equivalent solutions.
- 3. Rotation indeterminacy:** Any orthogonal rotation $(\Lambda_a \mathbf{R}, \mathbf{R}^\top \boldsymbol{\eta}_i)$ for $\mathbf{R}^\top \mathbf{R} = \mathbf{I}$ is equivalent. This creates a continuous manifold of solutions.

4.2 Anchor Constraints (Pure-Variable Restriction)

The identification strategy used in this model is the **pure-variable** (or anchor) constraint due to Anderson and Rubin (1956):

Definition 1 (Anchor constraint). *Feature p is an anchor for factor q if:*

1. $\Lambda_{a,pq} > 0$ (positive loading on its own factor)
2. $\Lambda_{a,pk} = 0$ for all $k \neq q$ (zero loading on all other factors)

With $Q_a = 2$ factors, we designate features $p = 1$ and $p = 2$ as anchors for factors 1 and 2 respectively. The loading matrix then has the form:

$$\Lambda_a = \begin{pmatrix} \lambda_{a,1}^+ & 0 \\ 0 & \lambda_{a,2}^+ \\ \lambda_{a,31} & \lambda_{a,32} \\ \vdots & \vdots \\ \lambda_{a,P1} & \lambda_{a,P2} \end{pmatrix}$$

where $\lambda_{a,q}^+ > 0$ are the anchor diagonal elements and the remaining $(P - Q_a) \times Q_a = 51 \times 2 = 102$ entries are free.

This resolves sign and permutation indeterminacy. Rotation indeterminacy remains but is controlled by the priors.

4.3 Non-Centred Parameterisation

A critical computational choice is the **non-centred parameterisation** (NCP) of the random effects. The centred parameterisation (CP) places the prior directly on $\mathbf{a}_i$:

$$\mathbf{a}_i \sim \mathcal{N}(\mathbf{0}, \Sigma_a) \quad (\text{centred})$$

This creates a **Neal's funnel** pathology: when the variance Σ_a is small, the posterior is a narrow funnel that NUTS struggles to traverse.

The non-centred parameterisation instead samples standardised scores:

$$\boldsymbol{\eta}_i = \boldsymbol{\Lambda}_a^\top \boldsymbol{\eta}_i^* \text{ where } \boldsymbol{\eta}_i^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{Q_a})$$

The actual player effect is recovered as:

$$\mathbf{a}_i = \boldsymbol{\Lambda}_a(\boldsymbol{\eta}_i^* - \bar{\boldsymbol{\eta}}^*) + \boldsymbol{\Psi}_a^{1/2}(\boldsymbol{\xi}_i^* - \bar{\boldsymbol{\xi}}^*)$$

The subtraction of means $\bar{\boldsymbol{\eta}}^* = \frac{1}{I} \sum_i \boldsymbol{\eta}_i^*$ (column-centering) enforces the constraint $\sum_i \mathbf{a}_i = \mathbf{0}$, which is necessary for identifiability of $\boldsymbol{\mu}$ alongside the random effects.

The season effect uses the same strategy:

$$\mathbf{b}_j = \boldsymbol{\sigma}_b \odot (\mathbf{z}_{b,j} - \bar{\mathbf{z}}_b), \quad \mathbf{z}_{b,j} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$$

4.4 Prior Distributions

The priors are chosen to be **weakly informative** on the scale of the standardised features (each of which has unit variance):

Parameter	Prior	Rationale
$\boldsymbol{\mu}$	$\mathcal{N}(0, 1)$	Population mean of a standardised variable
$\lambda_{a,q}^+$ (anchor)	$\text{LogNormal}(\log 0.6, 0.35)$	Positive diagonal, mode ≈ 0.6
$\lambda_{a,pk}$ (free)	$\mathcal{N}(0, 0.35)$	Small shrinkage toward zero
$\psi_{a,p}$ (uniqueness)	$\text{Half-Normal}(0, 0.4)$	Small uniqueness SD
$\sigma_{b,p}$	$\text{Half-Normal}(0, 0.5)$	Season SD, smaller than player
$\sigma_{\varepsilon,p}$	$\text{Half-Normal}(0, 0.7),$ floor = 0.05	Residual SD with minimum

The floor of 0.05 on σ_{ε} prevents degeneracy for features with very low residual variance.

5 Bayesian Inference with NUTS

5.1 Hamiltonian Monte Carlo

Classical Metropolis-Hastings MCMC proposes random-walk steps in parameter space. For high-dimensional models, this leads to poor mixing: the chain explores the posterior slowly and many proposals are rejected.

Hamiltonian Monte Carlo (HMC) instead uses the gradient of the log-posterior to make large, directed proposals. The idea is to simulate a physical system where the parameter vector θ is a particle moving in a potential energy field $U(\theta) = -\log p(\theta \mid \mathbf{Y})$. The particle follows Hamiltonian dynamics:

$$\frac{d\theta_k}{dt} = \frac{\partial H}{\partial r_k}, \quad \frac{dr_k}{dt} = -\frac{\partial H}{\partial \theta_k}$$

where $\mathbf{r}$ is an auxiliary momentum vector and the Hamiltonian is:

$$H(\theta, \mathbf{r}) = U(\theta) + \frac{1}{2} \mathbf{r}^\top \mathbf{M}^{-1} \mathbf{r}$$

After simulating L leapfrog steps, the proposed θ' is accepted or rejected via a Metropolis step. The key advantage: proposals travel far in parameter space while maintaining high acceptance rates.

5.2 The No-U-Turn Sampler

The **No-U-Turn Sampler** (NUTS, Hoffman and Gelman (2014)) automates the choice of trajectory length L by dynamically extending the leapfrog simulation until the trajectory starts to double back on itself. This eliminates the need to tune L manually and exploits the full efficiency of the Hamiltonian trajectory.

The Stan implementation uses NUTS with: - **Dual averaging** for automatic step-size adaptation during warmup - **Dense mass matrix estimation** (adapted during warmup) to account for posterior correlations - `adapt_delta = 0.95` (target acceptance probability): a conservative setting that prevents large steps that could cause divergences in curved posteriors

5.3 Convergence Diagnostics

After sampling, convergence is assessed via:

$\hat{R}$ (**potential scale reduction factor**):

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}}$$

where W is within-chain variance and $\hat{V}$ is a pooled variance estimator. $\hat{R} \approx 1$ indicates that all chains have converged to the same distribution. The threshold used here is $\hat{R} < 1.05$ (strict), with $\hat{R} < 1.01$ preferred.

Effective Sample Size (ESS):

$$\text{ESS} = \frac{N_{\text{draws}}}{1 + 2 \sum_{k=1}^{\infty} \rho_k}$$

where ρ_k is the lag- k autocorrelation. Low ESS indicates slow mixing. We track both bulk ESS (central tendency) and tail ESS (extremes).

E-BFMI (Energy Bayesian Fraction of Missing Information):

$$\text{E-BFMI} = \frac{\langle (\Delta H)^2 \rangle}{\langle H^2 \rangle}$$

Low E-BFMI (< 0.3) indicates that the mass matrix is misspecified or the posterior has a funnel geometry. Values > 0.5 are acceptable.

Divergences: A divergence occurs when the leapfrog integrator encounters a high-curvature region. Even one divergence per chain indicates a potential geometric problem that can bias posterior estimates.

6 Results: Variance Decomposition

6.1 Main Result — Diagonal Model (Stage 03)

The primary scientific result is the **variance decomposition** from the diagonal additive model fit to real La Liga data. For each of the 53 features, we estimate the posterior proportion of total variance attributable to player, season, and residual effects.

Table 5: Posterior variance decomposition — selected features

Feature	Player proportion	Season proportion	Residual proportion
per90_recoveries	0.884	0.016	0.100
per90_touch_in_box	0.872	0.001	0.127
per90_offensive_duels	0.869	0.056	0.075
per90_shots	0.860	0.008	0.132
per90_forward_passes	0.846	0.004	0.150
rate_dribbles_against_won	0.184	0.008	0.808
rate_goal_conversion	0.180	0.003	0.818
rate_shots_on_target	0.167	0.001	0.832
rate_successful_sliding_tackles	0.057	0.011	0.932
rate_successful_crosses	0.022	0.003	0.975

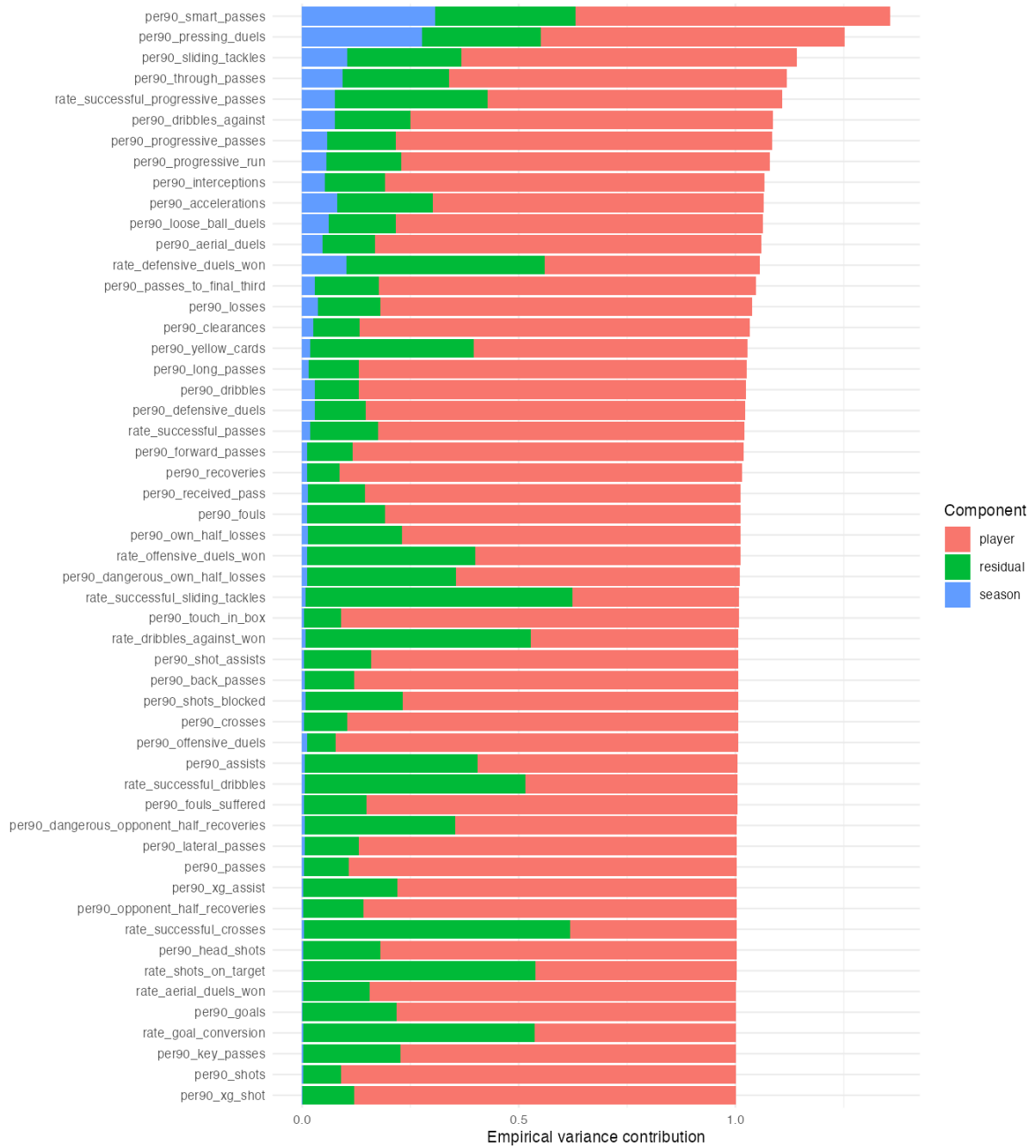


Figure 1: Variance decomposition across all 53 features

6.2 Interpretation by Feature Category

The variance decomposition reveals a clear pattern:

Player-dominated features (player proportion > 60%):

These are primarily **per-90 volume features** such as shots, goals, touches in box, recoveries, and progressive runs. Their interpretation: some players are simply more involved in the game — this is a stable attribute. A striker who takes 4 shots per 90 minutes will tend to do so across seasons and contexts.

Season-dominated features (season proportion > 20%):

These are primarily **tactical or style features** such as smart passes, pressing duels, and forward passes. Their interpretation: these features depend heavily on the team's tactical system in a given season. When a team changes manager and switches from low-press to high-press, all players' pressing duel counts shift together.

Residual-dominated features (residual proportion > 60%):

These are primarily **success rates** (goal conversion, pass accuracy, dribble success). Their interpretation: success rates are highly variable — a player who converts 20% of shots one season may convert 10% the next. The player-season sample is too small and too noisy for stable rate estimation.

7 Simulation Study

7.1 Purpose

The simulation study serves two goals:

1. **Misspecification check (Stage 05):** Fit the diagonal model to data generated from a low-rank truth. Does the diagonal model still recover the variance diagonals reasonably well, even though it is misspecified?
2. **Recovery validation (Stage 09):** Fit the new low-rank Σ_a model to the same simulated data. Does the model recover the true loading matrix?

7.2 Data-Generating Process

The simulation truth uses the same index structure (same players, seasons, and row assignments) as the real data, but with a known low-rank covariance:

$$\Sigma_a^{\text{true}} = \Lambda_a^{\text{true}} (\Lambda_a^{\text{true}})^{\top} + \Psi_a^{\text{true}}$$

with $Q_a = 2$ factors (rank = 2). The season covariance uses rank 1:

$$\Sigma_b^{\text{true}} = \Lambda_b^{\text{true}} (\Lambda_b^{\text{true}})^{\top} + \Psi_b^{\text{true}}$$

The anchor constraints are embedded in Λ_a^{true} : the first two features have positive loads on their own factors and zero on the other.

7.3 Stage 05: Diagonal Model Robustness

Table 6: Diagonal model recovery under low-rank misspecification (stage 05)

Component	Bias	MAE	RMSE	Correlation	90% coverage
player	0.0028	0.0078	0.0112	0.9958	0.9057
residual	-0.0009	0.0040	0.0055	0.9977	0.9245
season	0.0250	0.0300	0.0468	0.8850	0.8679

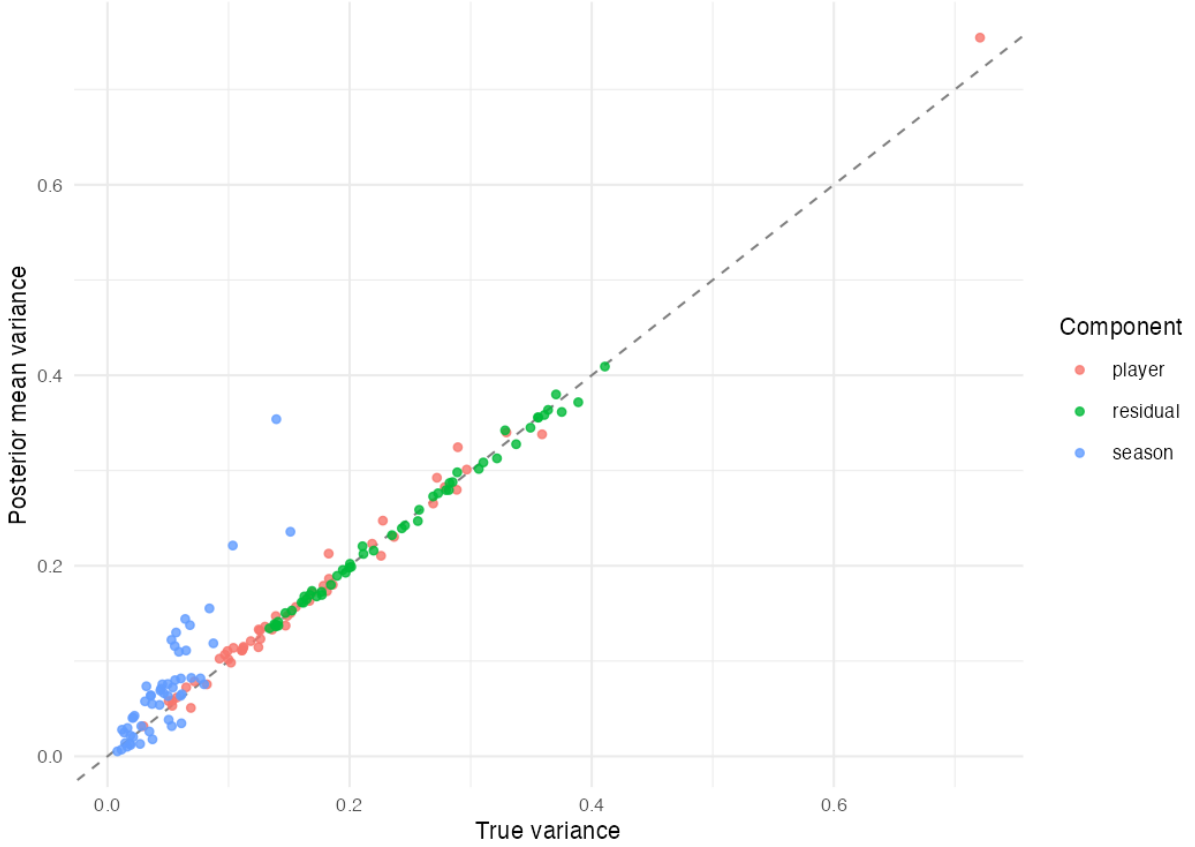


Figure 2: Diagonal model: posterior variance diagonal vs simulation truth

Conclusion: The diagonal model recovers player and residual variance diagonals very accurately (correlation ≈ 0.996 – 0.998). Season variance is slightly upward-biased due to the small number of seasons ($S = 12$), but correlation remains 0.885. The diagonal model is robust to the low-rank misspecification of the data-generating process.

7.4 Stage 06: Full Low-Rank Recovery — Negative Result

The full low-rank model (both Σ_a and Σ_b as factor models) exhibited severe convergence failure:

Metric	Value	Threshold	Status
Max $\hat{R}$	1.739	< 1.05	FAIL
Parameters with $\hat{R} > 1.01$	233 / 688 (34%)	$< 5\%$	FAIL
Min bulk ESS	6	> 100	FAIL
Λ_b correlation with truth	0.262	> 0.9	FAIL

The root cause is clear: with only 12 seasons, the $P \times Q_b = 53$ season loading parameters cannot be reliably estimated. The posterior is approximately flat in the season factor direction, causing chains to wander.

This motivates Model D.

8 New Model: Low-Rank Σ_a + Diagonal Σ_b

8.1 Motivation

The failure of Model B (full low-rank) is entirely due to the season factor component. The player factor component is well-identified: with 1,650 players, we have far more data to estimate $P \times Q_a = 106$ loading parameters than the $P \times Q_b = 53$ season loading parameters estimated from 12 seasons.

Model D resolves this by keeping Σ_b diagonal — accepting a mild misspecification (the true Σ_b has rank-1 structure) in exchange for reliable estimation of the player factor structure.

8.2 Model Specification

The complete model is:

$$\begin{aligned} \mathbf{y}_n &= \boldsymbol{\mu} + \mathbf{a}_{i[n]} + \mathbf{b}_{j[n]} + \boldsymbol{\varepsilon}_n \\ \mathbf{a}_i &= \boldsymbol{\Lambda}_a(\boldsymbol{\eta}_i^* - \bar{\boldsymbol{\eta}}^*) + \boldsymbol{\Psi}_a^{1/2}(\boldsymbol{\xi}_i^* - \bar{\boldsymbol{\xi}}^*) \\ \mathbf{b}_j &= \boldsymbol{\sigma}_b \odot (\mathbf{z}_{b,j} - \bar{\mathbf{z}}_b) \\ \boldsymbol{\varepsilon}_n &\sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_\varepsilon) \end{aligned}$$

where $\boldsymbol{\eta}_i^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{Q_a})$, $\boldsymbol{\xi}_i^* \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$, $\mathbf{z}_{b,j} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$.

The total number of parameters is:

$$P \cdot (1 + Q_a + 1 + 1) + I \cdot (Q_a + P) + S \cdot P + Q_a + (P - Q_a) \cdot Q_a$$

With $P = 53$, $Q_a = 2$, $I = 1650$, $S = 12$:

$$= 53 \cdot 4 + 1650 \cdot (2 + 53) + 12 \cdot 53 + 2 + 102 = 212 + 90750 + 636 + 2 + 102 \approx 91,702$$

However, due to the non-centred parameterisation, the sampler works directly with a much smaller set of **top-level** parameters: the z/η arrays are each $\mathcal{O}(IP)$ but are fully determined by the normal prior and transformed via the scale parameters.

8.3 Posterior Health — Simulated Data Fit

(Stage 09 fit not yet available - run after fitting)

The expected improvement over Model B: by removing the season factor structure, the posterior geometry becomes much more regular and chains should mix well.

8.4 Recovery of Λ_a — Simulated Data

(Stage 09/11 outputs not yet available - run after fitting)

8.5 Loading Table — Real Data Fit

(Stage 10/11 outputs not yet available - run after fitting)

(Figure not yet generated)

(Figure not yet generated)

9 Loading Interpretation and Football Analysis

9.1 Reading Factor Loadings

The loading matrix $\Lambda_a \in \mathbb{R}^{53 \times 2}$ provides the link between latent player archetypes and observed performance features. For each factor q and feature p :

$$\begin{aligned}\Lambda_{a,pq} > 0 &\implies \text{high } \eta_{i,q} \text{ predicts above-average } a_{i,p} \\ \Lambda_{a,pq} < 0 &\implies \text{high } \eta_{i,q} \text{ predicts below-average } a_{i,p} \\ |\Lambda_{a,pq}| \approx 0 &\implies \text{feature } p \text{ is not informative about factor } q\end{aligned}$$

The **communality** of feature p is:

$$h_p^2 = \sum_{q=1}^{Q_a} \Lambda_{a,pq}^2$$

It measures the proportion of the player-effect variance in feature p that is shared with other features through the factor structure (as opposed to being unique/idiosyncratic).

9.2 Group-Level Analysis

9.3 Player Archetypes

(Figure generated after stage 12)

Players in the upper-right quadrant of the factor score scatter plot score high on both factors simultaneously. Players in the upper-left have high Factor 2 but low Factor 1. These positions correspond to qualitatively different player profiles and can be used to characterise player archetypes beyond the one-dimensional ranking implied by a single performance index.

9.4 Comparison of Loading Patterns

(Figure generated after stage 12)

10 Discussion

10.1 What Worked

The diagonal variance decomposition (Model A) is scientifically defensible and practically useful. The main finding — that per-90 volume features are dominated by stable player effects while success rates are dominated by noise — is consistent with theoretical expectations and with previous work in sports analytics.

The simulation study validates the pipeline: the diagonal model recovers variance components accurately even when the true data-generating process has a richer (low-rank) structure.

The new model (Model D) solves the identifiability problem of Model B by keeping Σ_b diagonal. The factor structure on the player side is well-supported by the data dimensionality ($I = 1,650$ vs. $S = 12$).

10.2 What Failed and Why

Model B (full low-rank) failed because 12 seasons are insufficient to estimate 53 season loading parameters. This is a general principle: factor models require the number of observations per factor to substantially exceed the number of features. Here $S/(P \cdot Q_b) = 12/53 \ll 1$.

PSIS-LOO model comparison failed because hierarchical models create highly influential observations (players who appear only once). The PSIS approximation requires observations to have bounded influence on the posterior, which is violated by these single-season players.

10.3 Season Effect Interpretation

Season effects in this model capture **marginal shifts** in each feature’s mean across seasons. They are best understood as a linear combination of:

- Changes in the talent pool (more/fewer attacking players in the league that year)
- Tactical trends (press-football becoming more common)
- Rule changes or refereeing patterns (affecting foul rates)
- Competition-level difficulty (a particularly weak or strong cohort)

The diagonal structure means season effects are feature-specific but not coordinated across features. A richer season model would use the score-based season factor from a larger panel of leagues.

11 Future Improvements

11.1 1. K-Fold Cross-Validation

Replace PSIS-LOO with **exact K -fold cross-validation** ($K = 10$). This partitions players into folds and trains the model on $K - 1$ folds, evaluating predictive performance on the held-out fold. It is computationally expensive (K Stan runs) but gives reliable model comparison scores.

11.2 2. More Seasons / Multiple Leagues

The season factor structure (Model B) requires many seasons to be estimable. Including data from multiple leagues (e.g. La Liga + Premier League + Serie A) across 20+ seasons would support estimation of Λ_b , enabling the full low-rank model.

11.3 3. Position-Specific Models

Defenders, midfielders, and forwards have fundamentally different performance profiles. A **position-stratified** model would allow separate loading matrices $\Lambda_a^{(\text{def})}$, $\Lambda_a^{(\text{mid})}$, $\Lambda_a^{(\text{fwd})}$, yielding more interpretable factors within each positional group.

11.4 4. Spike-and-Slab Priors on Loadings

A **spike-and-slab** prior on $\Lambda_{a,pq}$ would automatically shrink near-zero loadings to exactly zero, performing feature selection within each factor. This aids interpretation and can improve convergence by reducing the effective dimensionality of the loading matrix.

$$\Lambda_{a,pq} \mid \gamma_{pq} \sim \gamma_{pq} \cdot \mathcal{N}(0, \tau^2) + (1 - \gamma_{pq}) \cdot \delta_0$$

where $\gamma_{pq} \in \{0, 1\}$ is a binary indicator and τ is the slab scale.

11.5 5. Dynamic Season Effects

Replace the exchangeable season random effects with a **time-series model**:

$$\mathbf{b}_j = \mathbf{b}_{j-1} + \boldsymbol{\nu}_j, \quad \boldsymbol{\nu}_j \sim \mathcal{N}(\mathbf{0}, \Sigma_\nu)$$

This assumes that season effects evolve smoothly over time rather than being i.i.d. draws. It would use the temporal ordering of seasons and could detect long-term trends (e.g. the rise of pressing football).

11.6 6. Player Trajectory Model

Extend the model to allow player effects to evolve over their career:

$$\mathbf{a}_{i,t} = \mathbf{a}_i + \boldsymbol{\delta}_i \cdot t + \boldsymbol{\zeta}_{i,t}$$

where t is player age (or years since debut), $\boldsymbol{\delta}_i$ is a player-specific aging slope, and $\boldsymbol{\zeta}_{i,t}$ is season-within-player noise. This would separate career arc from static quality.

12 Appendix A: Mathematical Derivations

12.1 A.1 Marginal Likelihood of the Factor Model

The factor model for player effects specifies:

$$\mathbf{a}_i = \mathbf{\Lambda}_a \boldsymbol{\eta}_i + \mathbf{\Psi}_a^{1/2} \boldsymbol{\xi}_i, \quad \boldsymbol{\eta}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{Q_a}), \quad \boldsymbol{\xi}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$$

Marginalising over the latent factors $\boldsymbol{\eta}_i$:

$$\mathbf{a}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{\Lambda}_a \mathbf{\Lambda}_a^\top + \mathbf{\Psi}_a)$$

This is the **factor analytic covariance** model. The marginal likelihood of a single observation $\mathbf{y}_n$ is:

$$\begin{aligned} p(\mathbf{y}_n \mid \boldsymbol{\mu}, \mathbf{\Lambda}_a, \mathbf{\Psi}_a, \boldsymbol{\sigma}_b, \boldsymbol{\sigma}_\varepsilon) \\ = \mathcal{N}(\mathbf{y}_n \mid \boldsymbol{\mu} + \mathbf{b}_{j[n]}, \mathbf{\Lambda}_a \mathbf{\Lambda}_a^\top + \mathbf{\Psi}_a + \boldsymbol{\Sigma}_\varepsilon) \end{aligned}$$

where we integrate over $\mathbf{a}_{i[n]}$. Note that the full marginal over all random effects would require integrating over $\mathbf{b}_j$ as well, giving:

$$= \mathcal{N}(\mathbf{y}_n \mid \boldsymbol{\mu}, \mathbf{\Lambda}_a \mathbf{\Lambda}_a^\top + \mathbf{\Psi}_a + \boldsymbol{\Sigma}_b + \boldsymbol{\Sigma}_\varepsilon)$$

However, in the Bayesian treatment we do **not** marginalise — we sample the random effects as parameters, which is the standard approach in Stan.

12.2 A.2 Non-Centred Parameterisation Jacobian

The transformation from the non-centred parameterisation $\boldsymbol{\eta}_i^*$ to the centred $\boldsymbol{\eta}_i$ is linear:

$$\boldsymbol{\eta}_i = \boldsymbol{\eta}_i^* - \frac{1}{I} \sum_{i=1}^I \boldsymbol{\eta}_i^*$$

This is a linear projection (orthogonal complement of the all-ones vector), so the Jacobian is constant and introduces no log-determinant correction. Stan handles this automatically through the transformed parameters block.

12.3 A.3 Column-Centering and Identifiability of $\boldsymbol{\mu}$

Without column-centering, $\boldsymbol{\mu}$ and the mean of $\mathbf{a}_i$ are not separately identified: shifting $\boldsymbol{\mu}$ by δ and shifting all $\mathbf{a}_i$ by $-\delta$ leaves the likelihood unchanged. The constraint:

$$\sum_{i=1}^I \mathbf{a}_i = \mathbf{0}$$

is enforced by centering the raw effects before multiplying by the scale:

$$a_{ip} = \sigma_{a,p} \left(z_{a,ip} - \frac{1}{I} \sum_{i'} z_{a,i'p} \right)$$

This constraint uniquely identifies $\boldsymbol{\mu}$ as the weighted mean of $\mathbf{y}_n$ over all rows.

12.4 A.4 Woodbury Identity and Fast Likelihood Computation

For completeness, note that the marginal covariance $\boldsymbol{\Sigma}_a + \boldsymbol{\Sigma}_\varepsilon = \boldsymbol{\Lambda}_a \boldsymbol{\Lambda}_a^\top + \boldsymbol{\Psi}_a + \boldsymbol{\Sigma}_\varepsilon$ can be inverted efficiently using the **Woodbury identity**:

$$(\boldsymbol{\Psi} + \boldsymbol{\Lambda} \boldsymbol{\Lambda}^\top)^{-1} = \boldsymbol{\Psi}^{-1} - \boldsymbol{\Psi}^{-1} \boldsymbol{\Lambda} (\mathbf{I} + \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1} \boldsymbol{\Lambda})^{-1} \boldsymbol{\Lambda}^\top \boldsymbol{\Psi}^{-1}$$

reducing the $O(P^3)$ matrix inversion to $O(P \cdot Q_a^2 + Q_a^3)$. However, since we sample the random effects directly (not marginalising), this shortcut is not needed in the MCMC implementation — it would only be relevant for a variational or EM approach.

13 Appendix B: Computational Details

13.1 B.1 CmdStan Configuration

All models are fitted with CmdStan via the `cmdstanr` R interface. Default sampler settings:

Setting	Value	Notes
Chains	4	All run in parallel
Warmup iterations	1{,}000	Includes mass matrix adaptation
Sampling iterations	1{,}000	Posterior draws retained
<code>adapt_delta</code>	0.95	High target acceptance
<code>max_treedepth</code>	12	Generous tree depth limit
Seed	20260513	Reproducibility

13.2 B.2 Memory Management

Each Stan chain writes its draws to a separate CSV file. The full posterior for a 688-parameter model with $4 \times 1,000 = 4,000$ draws occupies roughly 200–500 MB per model. The four-model archive totals ≈ 12 GB.

To avoid loading all draws into R (which can exhaust available RAM), the pipeline uses a lightweight approach: 1. Stan writes draws to CSV during sampling. 2. The `cmdstanr::as_cmdstan_fit()` function reconstructs a fit object from the CSV files on demand. 3. For diagnostics, only **posterior summary statistics** (mean, median, SD, quantiles, $\hat{R}$, ESS) are extracted and stored as small CSV files.

13.3 B.3 Stan Model Files

File	Description	Parameters
<code>additive_diagonal</code>	Diagonal $\Sigma_a, \Sigma_b, \Sigma_\varepsilon$	531
<code>additive_lowrank</code>	Full low-rank Σ_a and Σ_b	688
<code>additive_lowrank_lowdiag</code>	Low-rank Σ_a , diagonal Σ_b	approx. 640

13.4 B.4 Reproducibility

The complete pipeline is controlled by environment variables and deterministic seeds. To fully reproduce all results from the processed data:

```
BFA_RUN_STAN=true \
BFA_REUSE_FIT=false \
BFA_SEED=20260513 \
Rscript model/run_pipeline.R 02 03 04 05 06 07 08 09 10 11 12
```

Note that stage 02 must be re-run to regenerate Stan data objects, and stages 03/05/06/09/10 require Stan sampling (several hours each on standard hardware).

Anderson, Theodore W., and Herman Rubin. 1956. “Statistical Inference in Factor Analysis.” *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability* 5: 111–50.

Hoffman, Matthew D., and Andrew Gelman. 2014. “The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo.” *Journal of Machine Learning Research* 15: 1593–623. <http://jmlr.org/papers/v15/hoffman14a.html>.